

**Task:** Look for a fake new that you would like to test.

# **FakeScope: Intelligent Fake News Detector**

*Why did the fake news article go to therapy?*

*Because it had too many issues and nobody believed its story!*



## Anti-LGBT demonstrators picket 'Pizzagate' restaurant on Inauguration Day

What the 'is Trump dead?' rumours reveal about our current moment

Vance says 'left-wing extremism' helped lead to Charlie Kirk's killing

## Why is this important?

¿Podemos Curar El Coronavirus Usando Paracetamol Y Lejía?

## Tackling coronavirus disinformation

It spreads misinformation, distort reality and manipulate public opinion.

It undermines trust in legitimate news sources and institutions.

Radio pionero en España y Portugal sitúa a Vox como el principal difusor de contenidos falsos

In Japan, a manga artist has predicted a disaster in early July, causing a drop in tourism

Infox transphobe sur Brigitte Macron: l'appel ne valide pas la rumeur, malgré les deux femmes l'ayant propagée

La cour d'appel de Paris a relaxé deux femmes, mais la justice rappelle que cette décision ne valide pas les rumeurs visant Brigitte Macron, contrairement à ce qu'affirment certains soutiens.

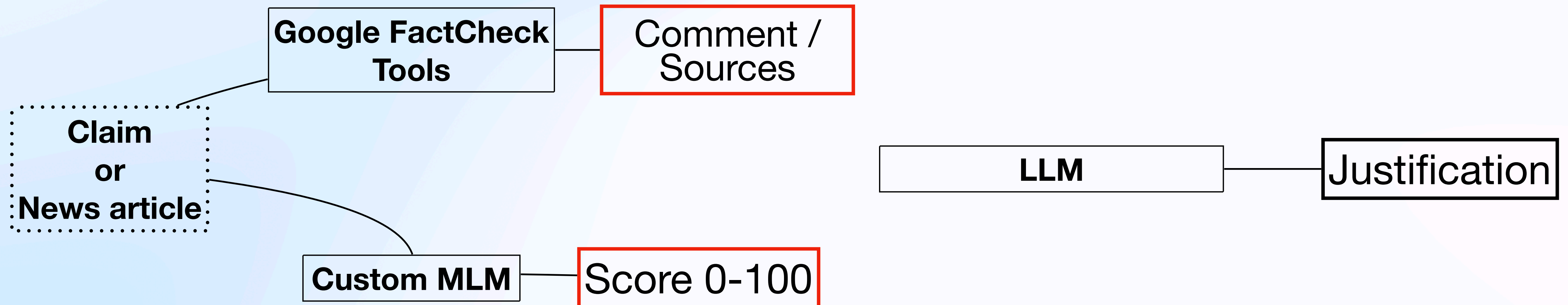
Le Monde avec AFP

Publié le 24 juillet 2025 à 17h40, modifié le 24 juillet 2025 à 18h52 • Lecture 1 min.

# GOAL

Evaluate the truthfulness of a news article by comparing it with a reliable online fact checker and assign a credibility score of 0-100, accompanied by an automatically generated brief justification.

It combines NLP, automated fact-checking, and semantic evaluation.





# Part I Data Pipeline and Preprocessing



# First Dataset

- From Kaggle:
  - World news and politics scrapped from Reuters and AP from 2016 to circa 2017 presidential elections.
  - Binary class: True/False
  - Title/Text/Class
    - *Subject/Date*
  - ((23481, 5), (21417, 5)) -> balanced.

It provides the model with rich context and deep semantic structures to learn from. The base of the training for the model.

**Donald Trump Sends Out Embarrassing New Year's Eve Message; This is Disturbing,** "Donald Trump just couldn't wish all Americans a Happy New Year and leave it at that. Instead, he had to give a shout out to his enemies, haters and the very dishonest fake news media. The former reality show star had just one job to do and he couldn't do it. As our Country rapidly grows stronger and smarter, I want to wish all of my friends, supporters, enemies, haters, and even the very dishonest Fake News Media, a Happy and Healthy New Year, President Angry Pants tweeted. 2018 will be a great year for America! As our Country rapidly grows stronger and smarter, I want to wish all of my friends, supporters, enemies, haters, and even the very dishonest Fake News Media, a Happy and Healthy New Year. 2018 will be a great year for America! Donald J. Trump (@realDonaldTrump) December 31, 2017 Trump's tweet went down about as well as you'd expect. What kind of president sends a New Year's greeting like this despicable, petty, infantile gibberish? Only Trump! His lack of decency won't even allow him to rise above the gutter long enough to wish the American citizens a happy new year! Bishop Talbert Swan (@TalbertSwan) December 31, 2017 no one likes you Calvin (@calvinstowell) December 31, 2017 Your impeachment would make 2018 a great year for America, but I'll also accept regaining control of Congress. Miranda Yaver (@mirandayaver) December 31, 2017 Do you hear yourself talk? When you have to include that many people that hate you you have to wonder? Why do they all hate me? Alan Sandoval (@AlanSandoval13) December 31, 2017 Who uses the word Haters in a New Year's wish?? Marlene (@marlene399) December 31, 2017 You can't just say happy new year? Koren pollitt (@Korencarpenter) December 31, 2017 Here's Trump's New Year's Eve tweet from 2016. Happy New Year to all, including to my many enemies and those who have fought me and lost so badly they just don't know what to do. Love! Donald J. Trump (@realDonaldTrump) December 31, 2016 This is nothing new for Trump. He's been doing this for years. Trump has directed messages to his enemies and haters for New Year's, Easter, Thanksgiving, and the anniversary of 9/11. [pic.twitter.com/4FPAe2KypA](https://pic.twitter.com/4FPAe2KypA) Daniel Dale (@ddale8) December 31, 2017 Trump's holiday tweets are clearly not presidential. How long did he work at Hallmark before becoming President? Steven Goodine (@SGoodine) December 31, 2017 He's always been like this . . . the only difference is that in the last few years, his filter has been breaking down. Roy Schulze (@thbthttt) December 31,



# Second Dataset

- Alternative dataset from Kaggle (LIAR Dataset):
  - Short political statements and claims.
  - Multi class: *Half-true, Pants on Fire...*  
-> Binary class: True/False
  - Title/url/Source/Text/date/Class
  - 10000 -> almost balanced.

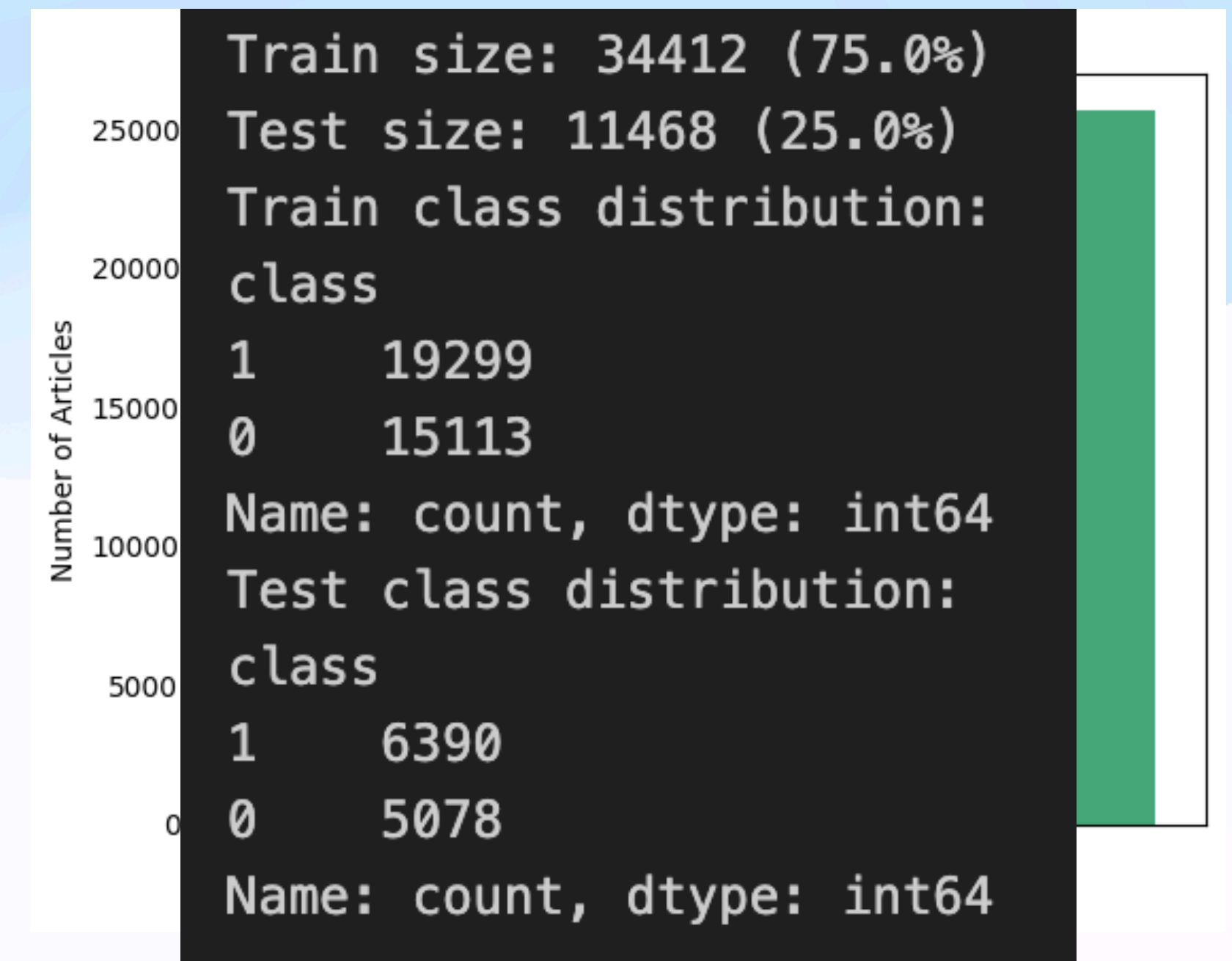
It helps the model detect short-form misinformation and specific false claims.

"Says Nate McMurray said that gun owners are part of the most insecure and dangerous club on earth, that the NRA are homegrown terrorists, and that abortion opponents are the American Taliban.",<https://www.politifact.com/factchecks/2020/jun/18/chris-jacobs/fact-checking-claims-chris-jacobs-nate-mcmurray-de/>,Chris Jacobs," June 13, 2020"," June 18, 2020",mostly-true"While in prison, boxer Jack Johnson invented and patented the first wrench, and white people insulted him by calling it a monkey wrench.",<https://www.politifact.com/factchecks/2020/jun/18/snoop-dogg/no-white-people-didnt-dub-black-boxer-jack-johnson/>,Snoop Dogg," June 15, 2020"," June 18, 2020",FALSE"According to its website Black Lives Matter, Inc is NOT a charity. It is a full-fledged corporation that does NOT have any locations.",<https://www.politifact.com/factchecks/2020/jun/17/candace-owens/how-black-lives-matter-global-network-set/>,Candace Owens," June 11, 2020"," June 17, 2020",FALSE"Donald Trump has opposed plans to clean up Toxic chemicals that cause cancer (that) are in the water . (and) put Wisconsin at risk.",<https://www.politifact.com/factchecks/2020/jun/12/league-conservation-voters/league-conservation-voters-ad-oversimplifies-trump/>,League of Conservation Voters," May 18, 2020 "," June 12, 2020",half-true""If Roe is overturned, Wisconsin has a law on the books that would immediately make abortion a crime.",<https://www.politifact.com/factchecks/2020/jun/11/planned-parenthood/yes-if-federal-abortion-law-were-overturned-wiscon/>,Planned Parenthood," March 4, 2020"," June 11, 2020",TRUE"Says the state health secretary says don't go to hospitals or clinics, now the tourism secretary says don't recreate.",<https://www.politifact.com/factchecks/2020/may/28/tom-tiffany/tiffany-wrong-claim-officials-dont-want-wisconsin/>,Tom Tiffany," May 26, 2020 "," May 28, 2020",barely-trueThis is 76 year old senile Joe Biden!!! That is a 15 year old girl with whom hes making out!!!,<https://www.politifact.com/factchecks/2020/may/28/facebook-posts/no-isnt-photo-biden-making-out-15-year-old-girl/>,Facebook posts," May 16, 2020 "," May 28, 2020",FALSE"Twitter is completely stifling FREE SPEECH by adding a fact-check label to Trumps tweet.",<https://www.politifact.com/factchecks/2020/may/27/donald-trump/trumps-pants-fire-claim-twitter-completely-stiflin/>,Donald Trump," May 26, 2020 "," May 27, 2020",pants-fireA photo



# Data Pipeline and Preprocessing

- Merging both datasets keeping only *text*, *title*, *class* columns.
- Removal of duplicates: 453 duplicates. No *nans*.
- Problem with encoding: *â€*, *Ã¢*... [library *ftfy*]
  - MD5 Content Hashing (Identifying Duplicates)
  - GroupShuffleSplit (Preventing Leakage)
- Stopwords (EN), Custom Domain Stopwords (sources, reporting terms, artifacts)



Environment setup, data loading with encoding fixes, text cleaning, and train/test splitting.



# Part II Feature Engineering



# Vectorising

**First training: overfitting**

**Hash-based deduplication to prevent train/test leakage.**

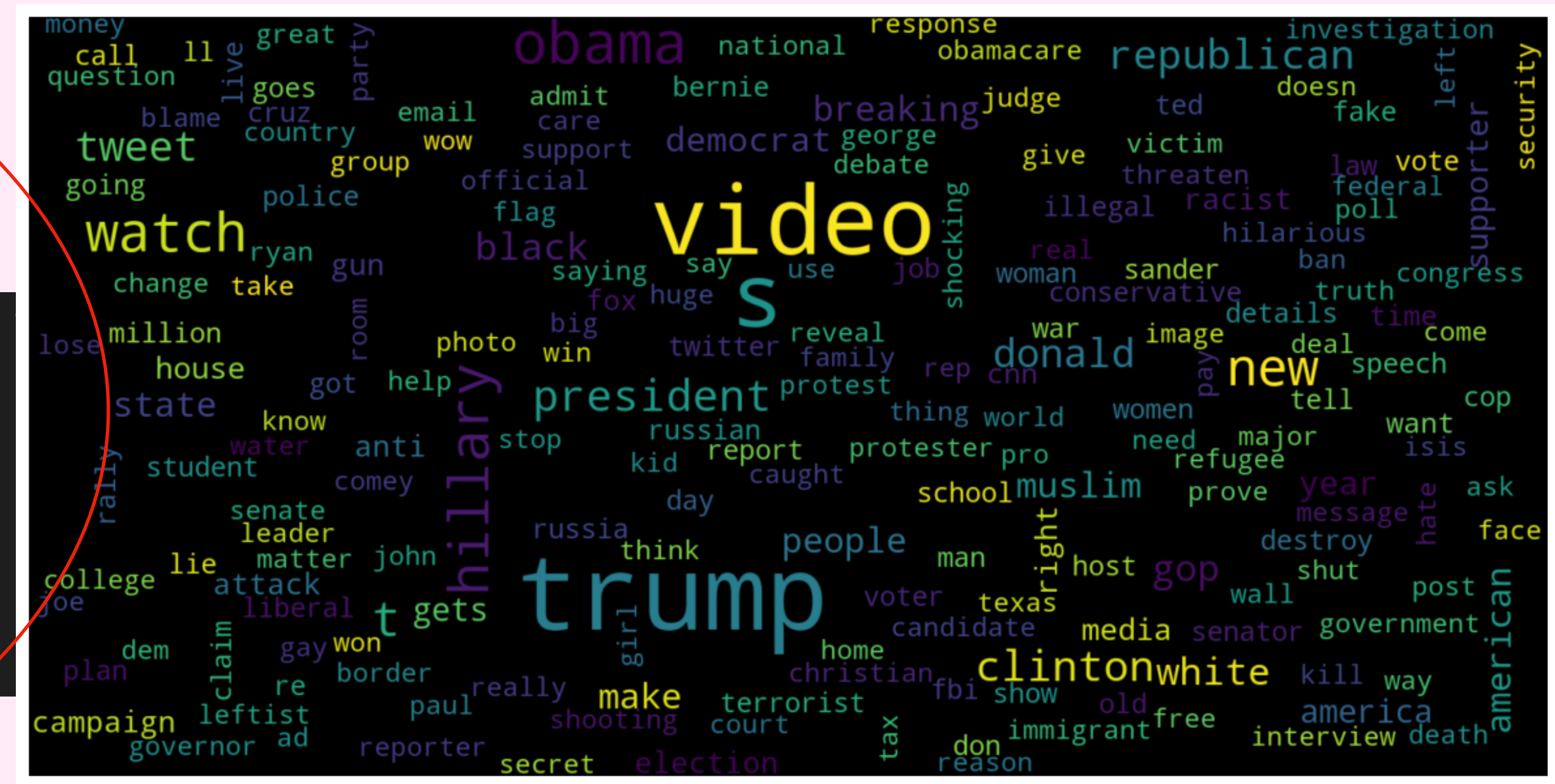
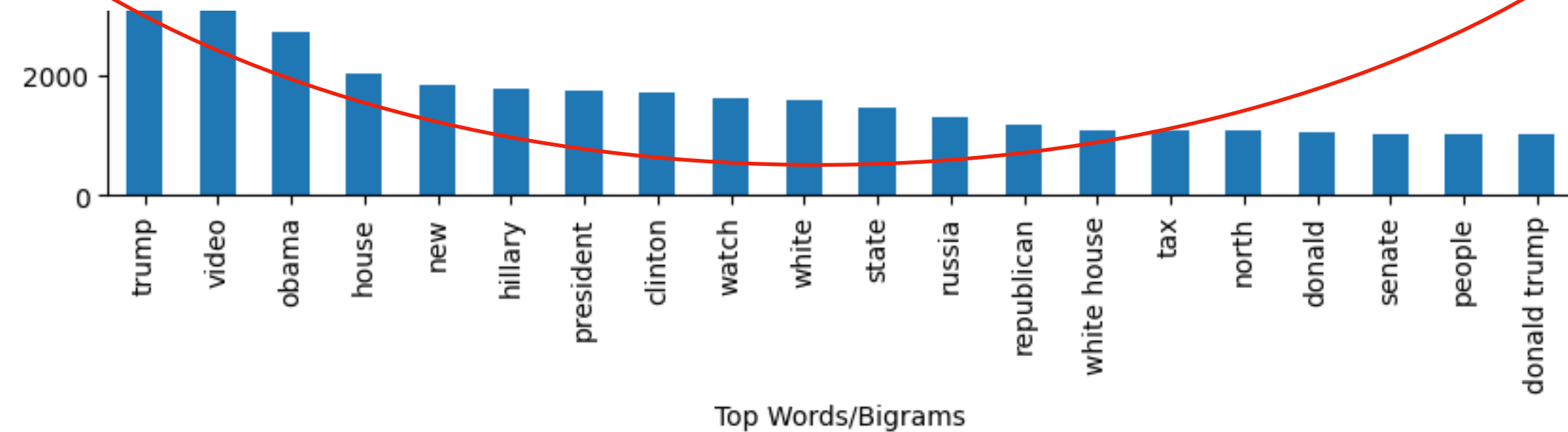
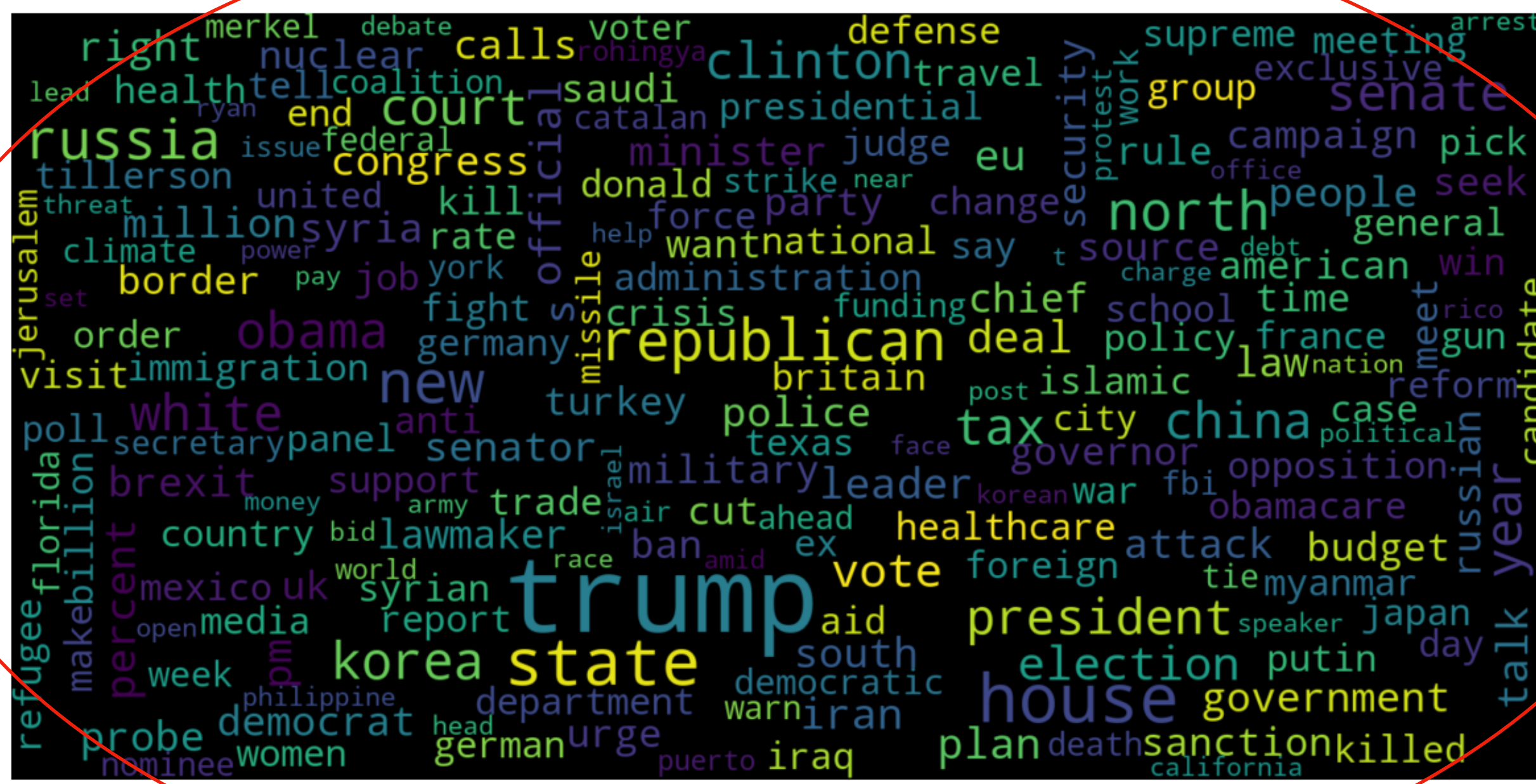
Strict TD-IDF Vectoriser:

- Max features: 3000.
- Personalised stop words list:  
'reuters', 'factbox', 'says',  
'associated', 'said', 'read', 'press'...
- MIN\_DF and MAX\_DF
- NGRAM\_RANGE (1,2)

**Ensuring 75/25 splitting.  
50% Fake and 50% True news, no bias.**



**Can you guess which one is the Fake word cloud?**





# Part III Baseline Models - Traditional ML



# Hyperparameter Tuning

## Avoiding overfitting

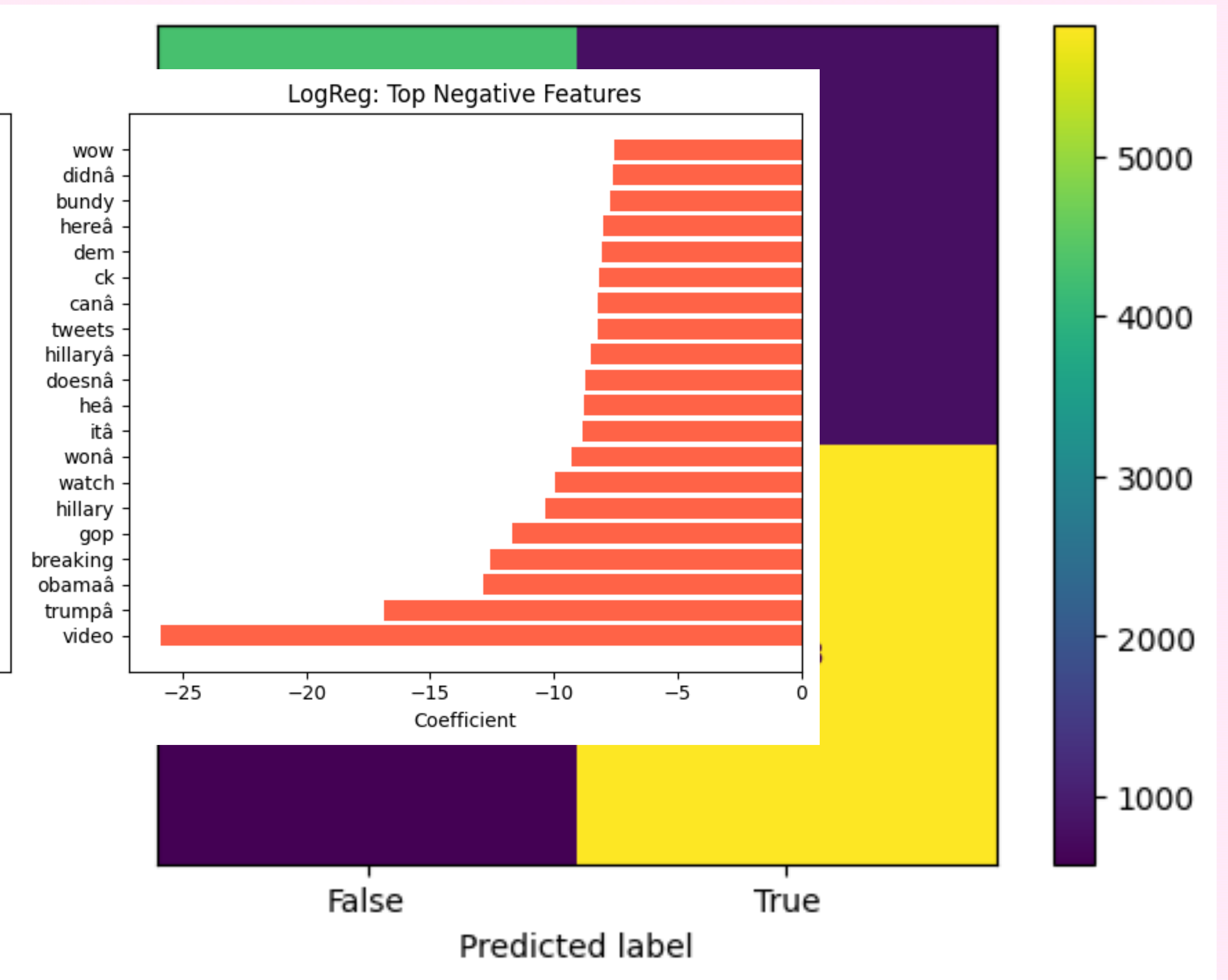
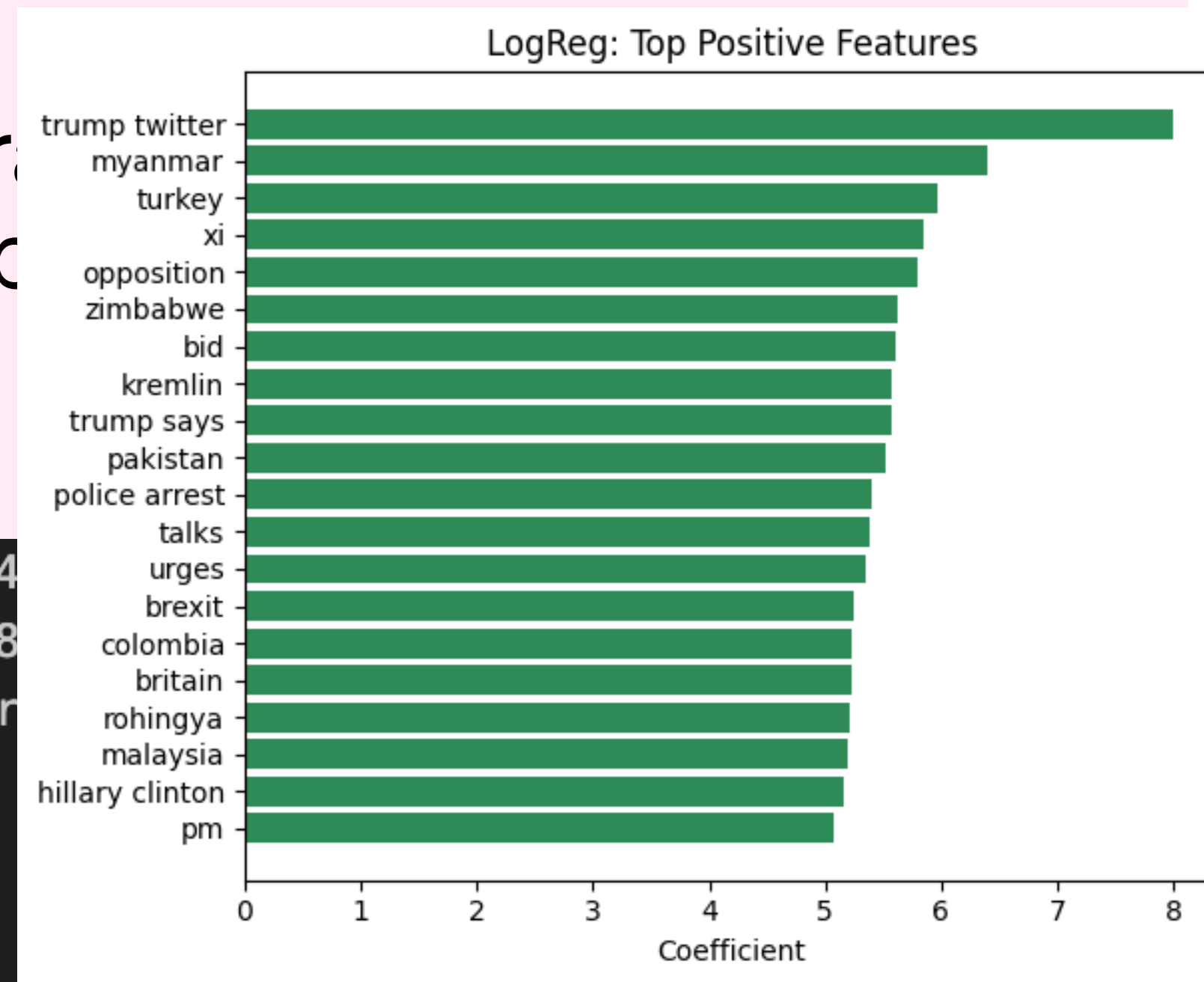
- Each model was tuned using GridSearchCV or RandomizedSearchCV.
- Models strong in generalisation and/or usually good with binary classifications: Logistic Regression, LightGBM, XGBoost, Decision Tree and Random Forest.
- FINAL GOAL: optimised ensemble approach.



# Logistic Regression

- 0.91308264558874  
0.88132193930938

| Variable | Coefficient |
|----------|-------------|
| pm       | 5.0         |



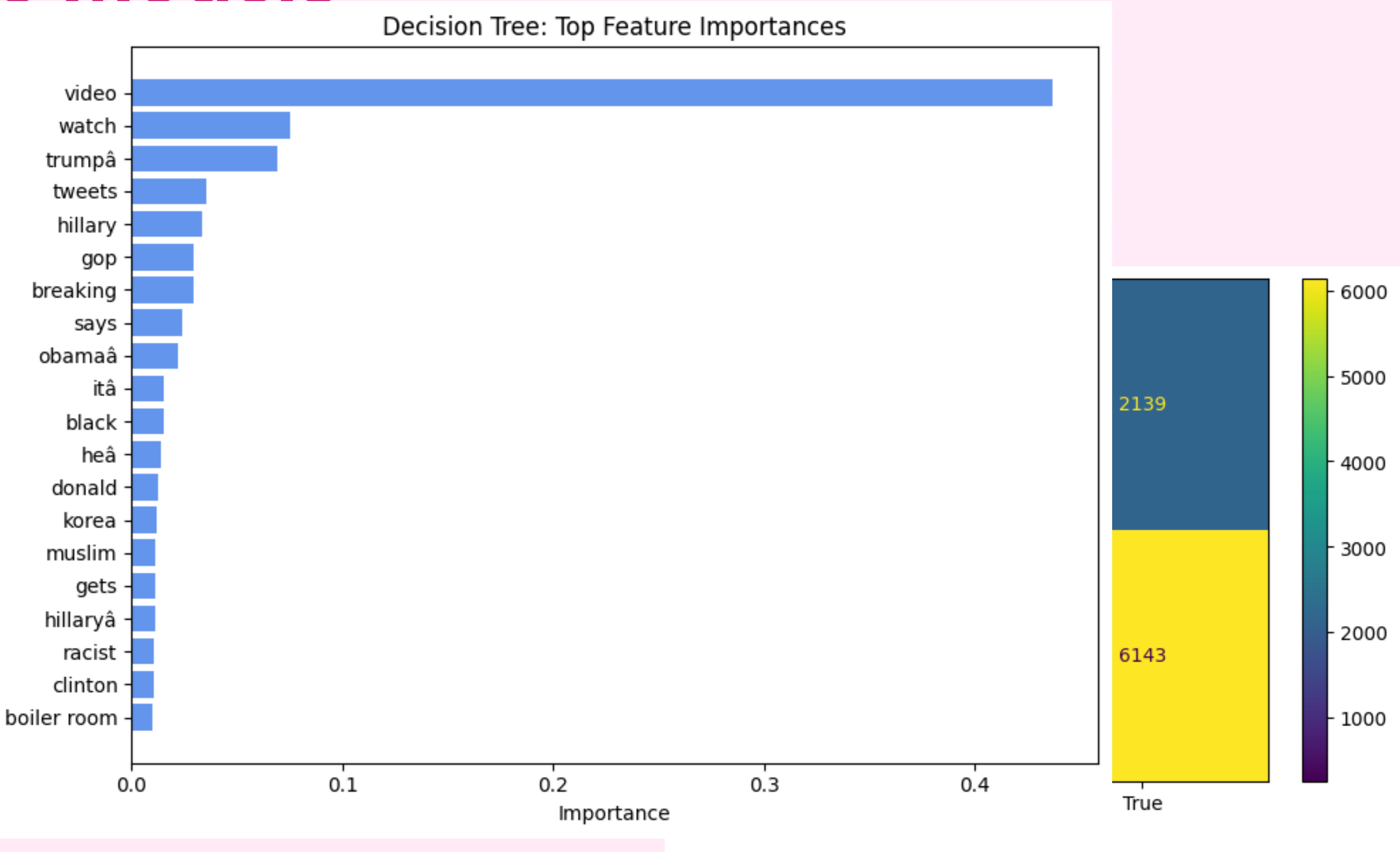


# Baseline Models

## Decision Tree

- Hyper-parameters
  - max\_depth

```
0.8141636638381
0.7919427973491
p
0
1
accuracy
macro avg
weighted avg
```





# Baseline

## Random Forest

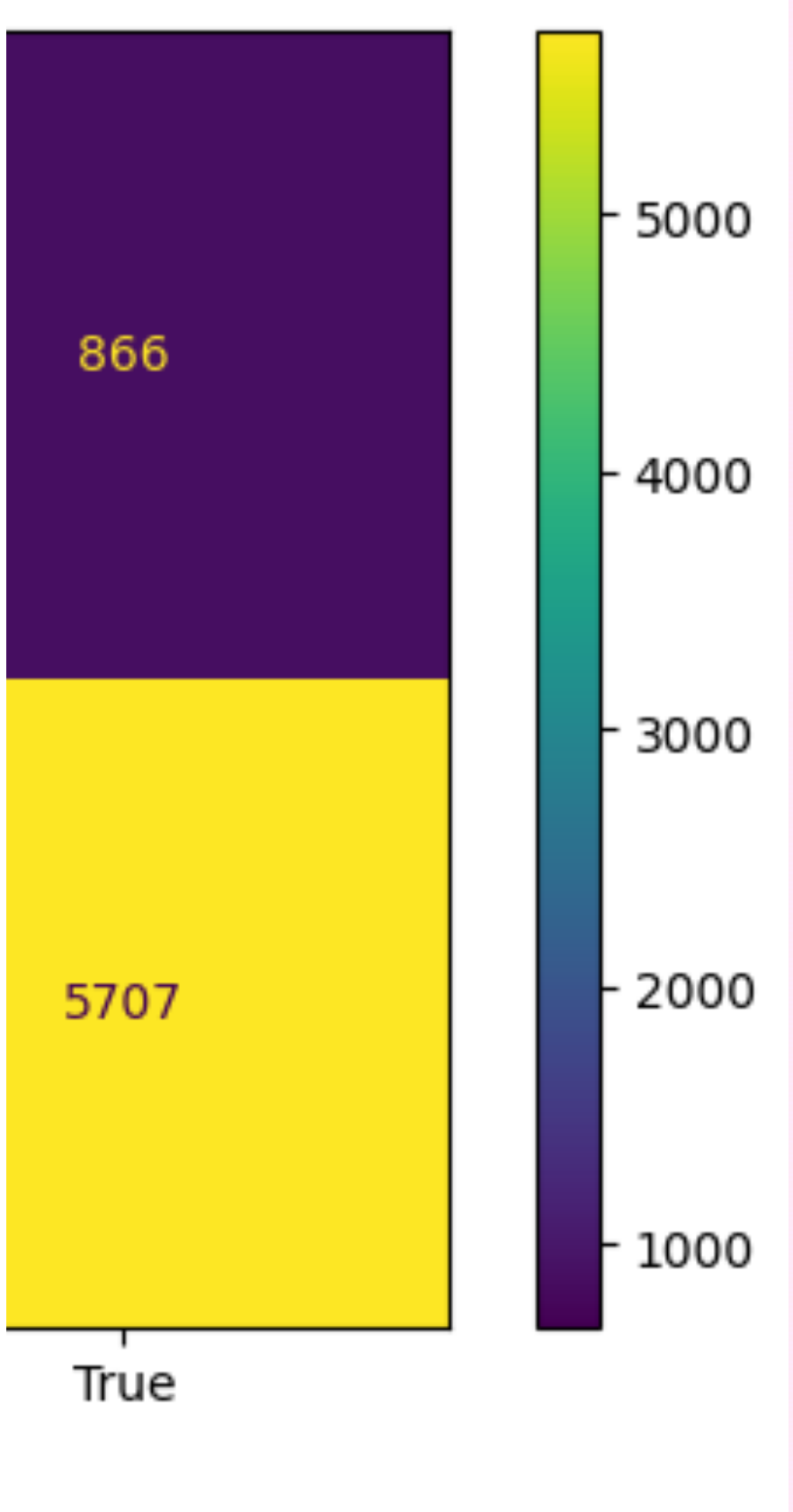
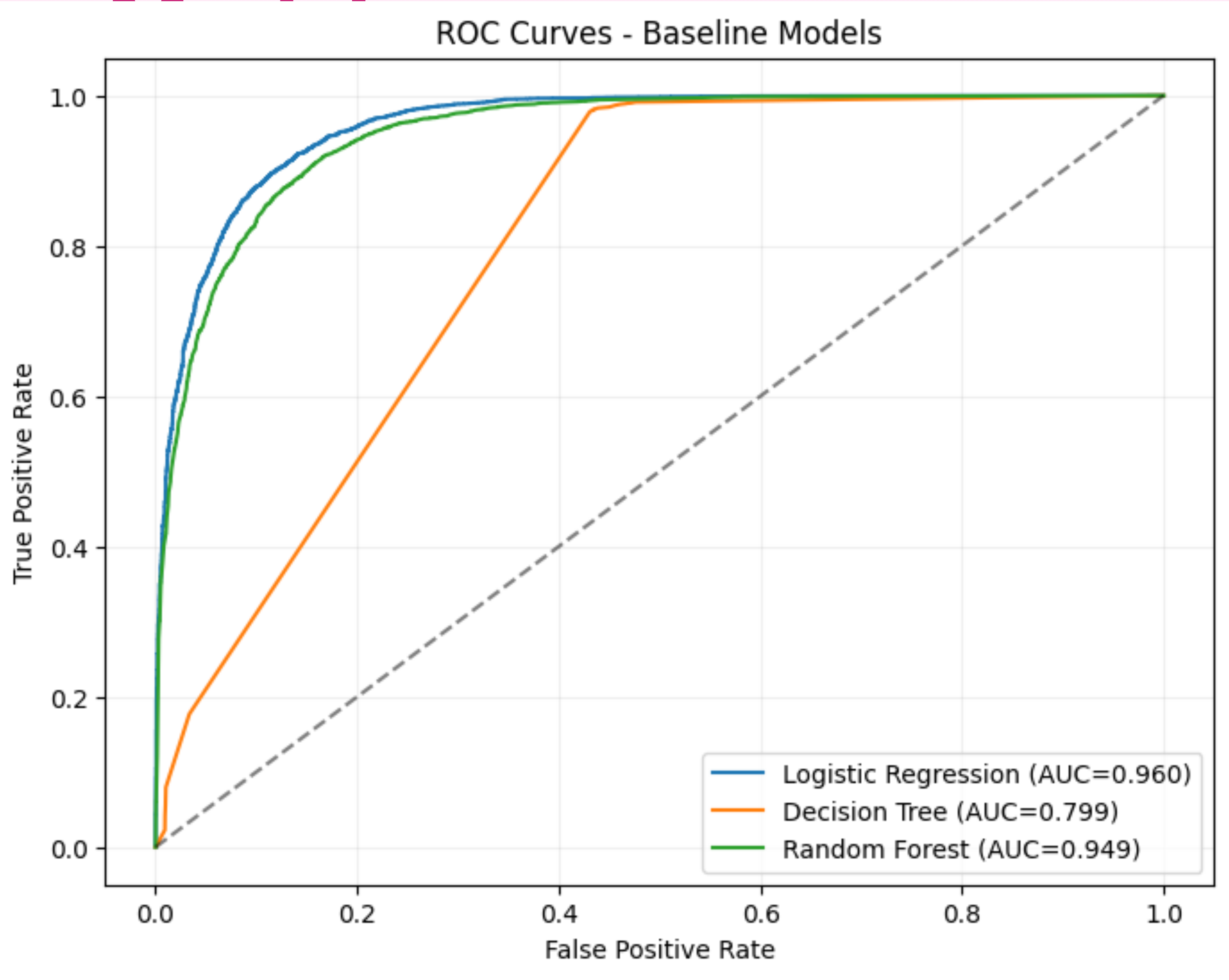
### Hyper-parameters

- 'max\_depth'
- 'min\_samples'
- 'n\_estimators'

Random Forest  
Random Forest  
Classification

accuracy  
macro avg  
weighted avg

0.86 0.86 0.86 11468



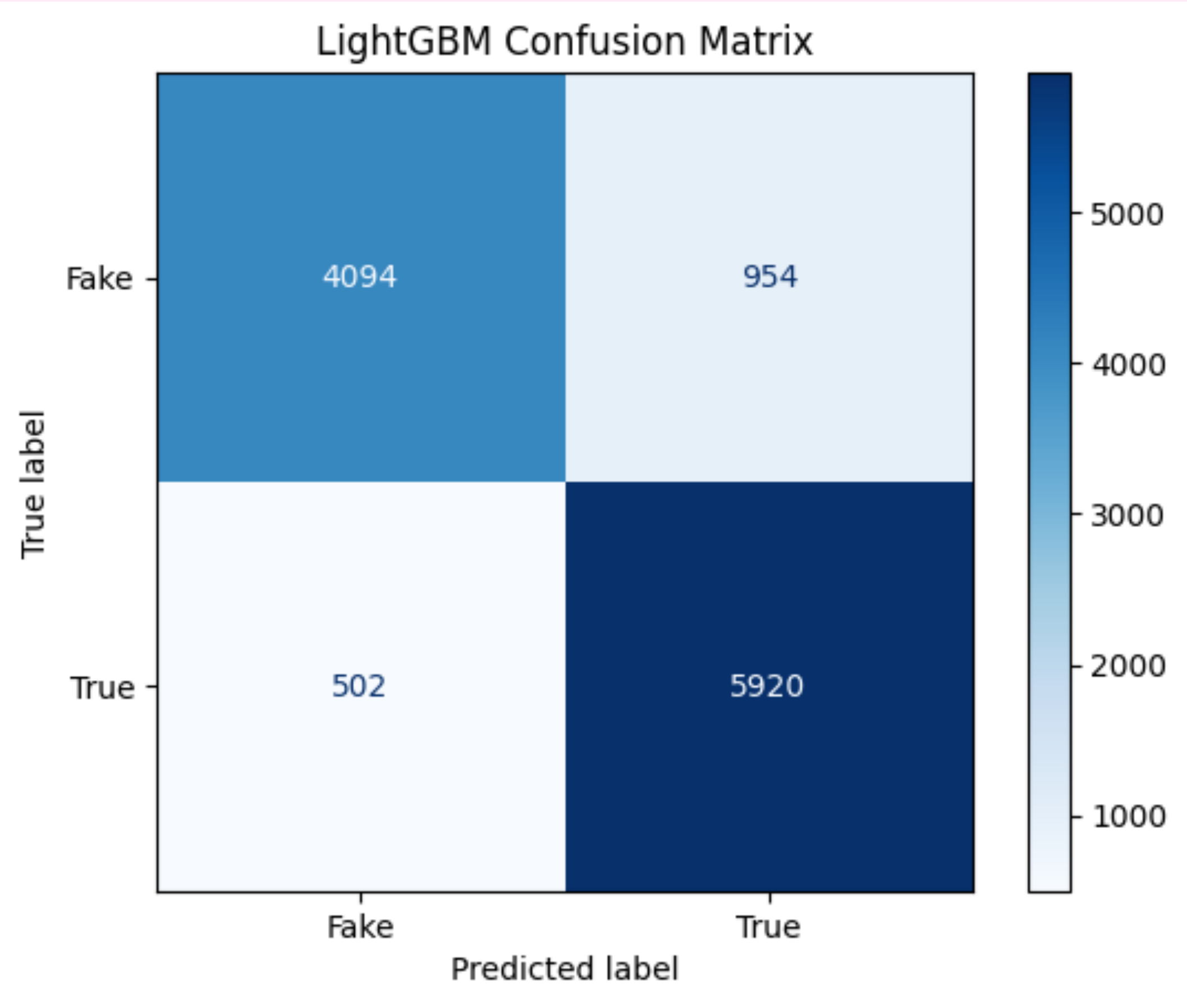


# Baseline Models

## LightGBM

Hyper-parameters: 'num\_leaves': 63,  
'feature\_fraction': 0.9,  
'bagging\_fraction': 0.8,  
'bagging\_freq': 5,  
'max\_depth': -1,

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.8908    | 0.8110 | 0.8490   | 5048    |
| 1            | 0.8612    | 0.9218 | 0.8905   | 6422    |
| accuracy     |           |        | 0.8731   | 11470   |
| macro avg    | 0.8760    | 0.8664 | 0.8698   | 11470   |
| weighted avg | 0.8742    | 0.8731 | 0.8722   | 11470   |



Leaf-Wise Growth

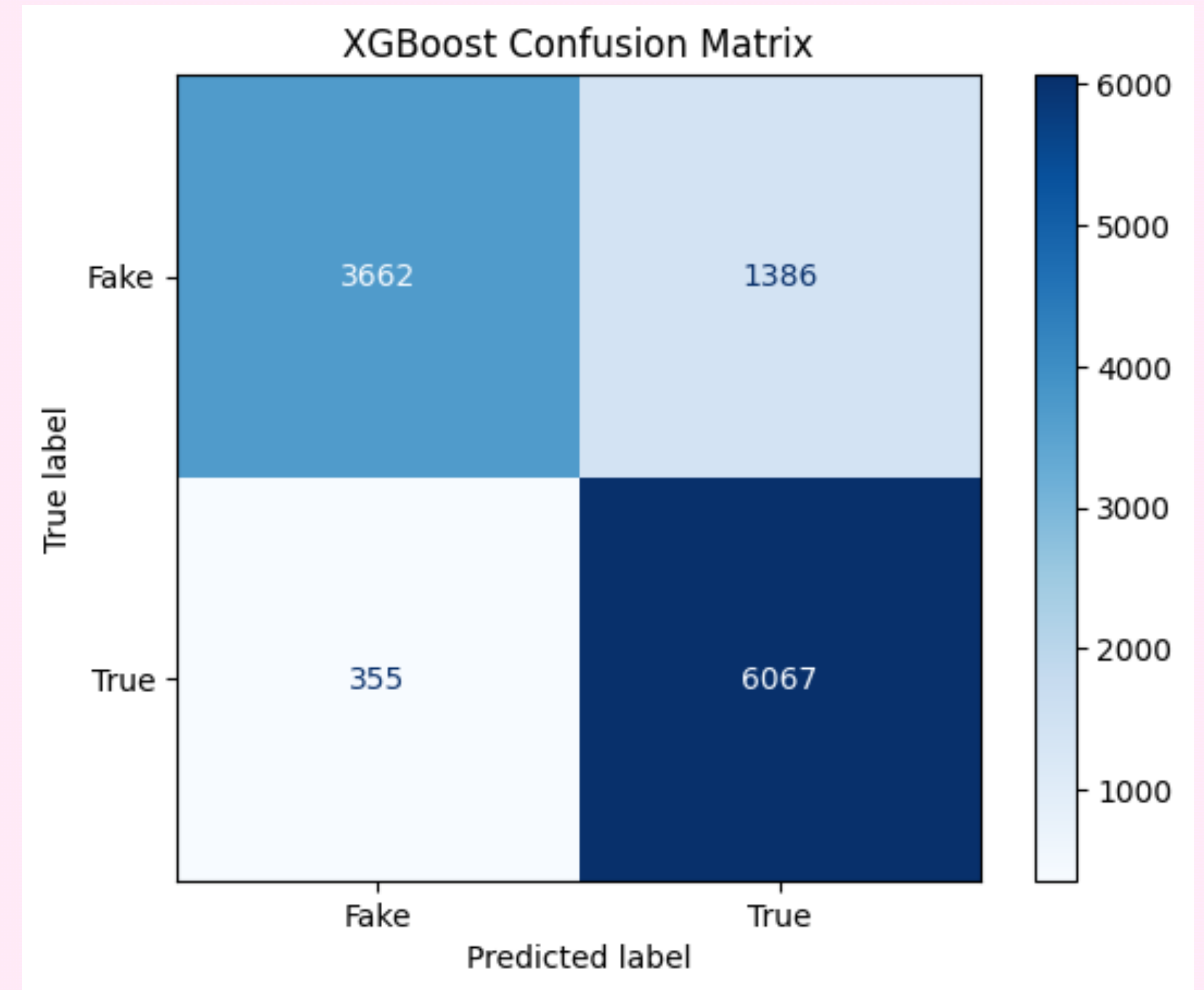


# Baseline Models

## XGBoost

- Hyper-parameters:
- ``n_estimators=200``
  - ``learning_rate=0.1``
  - ``max_depth=5``
  - ``colsample_bytree=0.7``
  - ``subsample=0.8``

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.9116    | 0.7254 | 0.8079   | 5048    |
| 1            | 0.8140    | 0.9447 | 0.8745   | 6422    |
| accuracy     |           |        | 0.8482   | 11470   |
| macro avg    | 0.8628    | 0.8351 | 0.8412   | 11470   |
| weighted avg | 0.8570    | 0.8482 | 0.8452   | 11470   |





| Model               | Accuracy | F1-Score (Macro) | Precision (Avg) | Recall (Avg) | Training Time |
|---------------------|----------|------------------|-----------------|--------------|---------------|
| LightGBM            | 87.87%   | 0.8757           | 0.8811          | 0.8727       | ~10 min       |
| Logistic Regression | 88.00%   | 0.88             | 0.88            | 0.88         | ~2 min        |
| Random Forest       | 86.49%   | 0.86             | 0.86            | 0.86         | ~15 min       |
| XGBoost             | 84.97%   | 0.8433           | 0.8621          | 0.8374       | ~20 min       |
| Decision Tree       | 79.00%   | 0.78             | 0.83            | 0.77         | <1 min        |

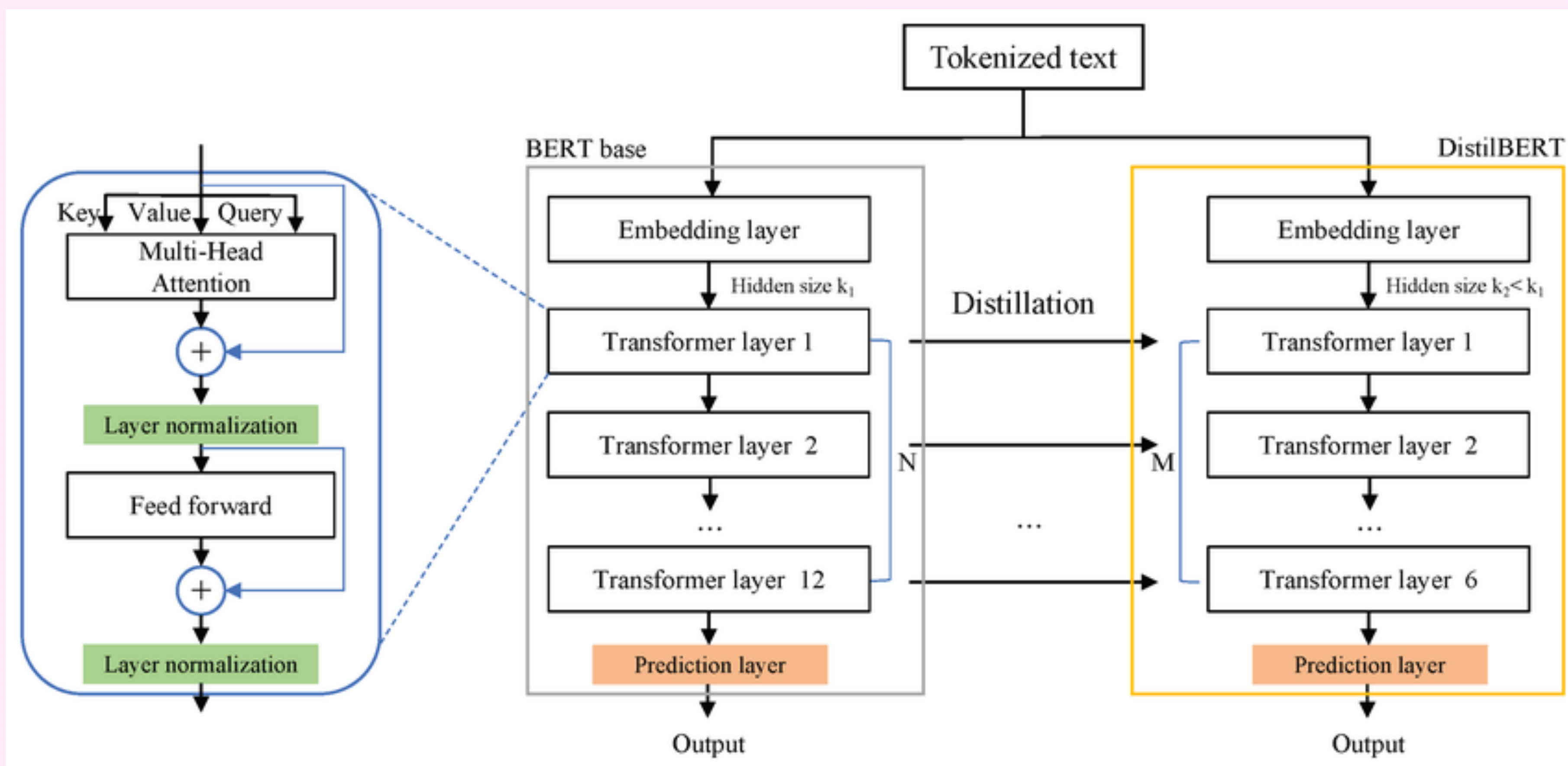


# Part IV Transformer Models



# Deep Learning Part

## DistilBERT: 1st phase of pre-training





# Deep Learning Part

## DistilBERT: 1st phase of pre-training

- Adaptability: Pre-training with Masked Language Modelling (MLM)

*Trump has opposed \_\_\_\_\_ to clean up toxic chemicals that \_\_\_\_\_ cancer.*

- Arguments:

```
output_dir='./mlm_results',
num_train_epochs=3, # Production-level: Should do 8 epochs (+48h) for good domain adaptation, but doing 3 for now
per_device_train_batch_size=16,
learning_rate=5e-5,
weight_decay=0.01,
save_strategy='epoch',
logging_steps=100,
fp16=False, # MPS (Apple Silicon) doesn't support fp16
use_mps_device=True, # Enable Apple Silicon GPU acceleration
```



# Deep Learning Part

## DistilBERT: 2nd phase of fine-tuning

- Unsupervised -> supervised training.

- Arguments:

```
output_dir='./results',  
eval_strategy='epoch',  
save_strategy='epoch',  
learning_rate=2e-5,  
per_device_train_batch_size=16,  
per_device_eval_batch_size=16,  
num_train_epochs=10,  
weight_decay=0.01,  
load_best_model_at_end=True,  
metric_for_best_model='accuracy',  
fp16=False,  
use_mps_device=True,  
logging_steps=50,  
save_total_limit=2,
```

| Epoch | Training Loss | Validation Loss | Accuracy | F1       |
|-------|---------------|-----------------|----------|----------|
| 1     | 0.100100      | 0.095068        | 0.954439 | 0.961210 |
| 2     | 0.106400      | 0.101621        | 0.956181 | 0.961810 |
| 3     | 0.083100      | 0.147827        | 0.954003 | 0.959860 |
| 4     | 0.030000      | 0.275460        | 0.948253 | 0.954343 |

**Stopped at epoch 4/10**

```
EarlyStoppingCallback(  
    early_stopping_patience=3,  
    early_stopping_threshold=0.01  
)
```



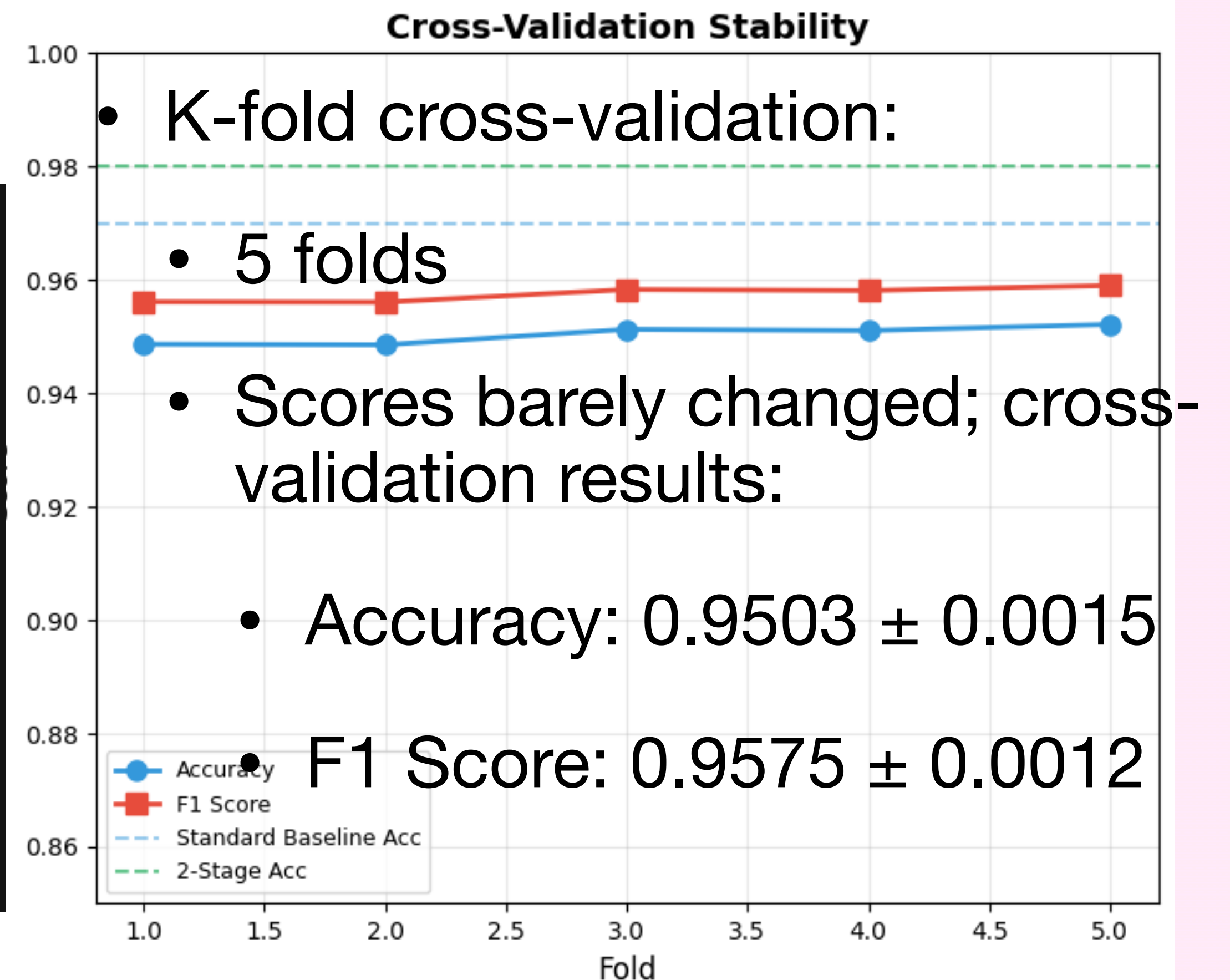
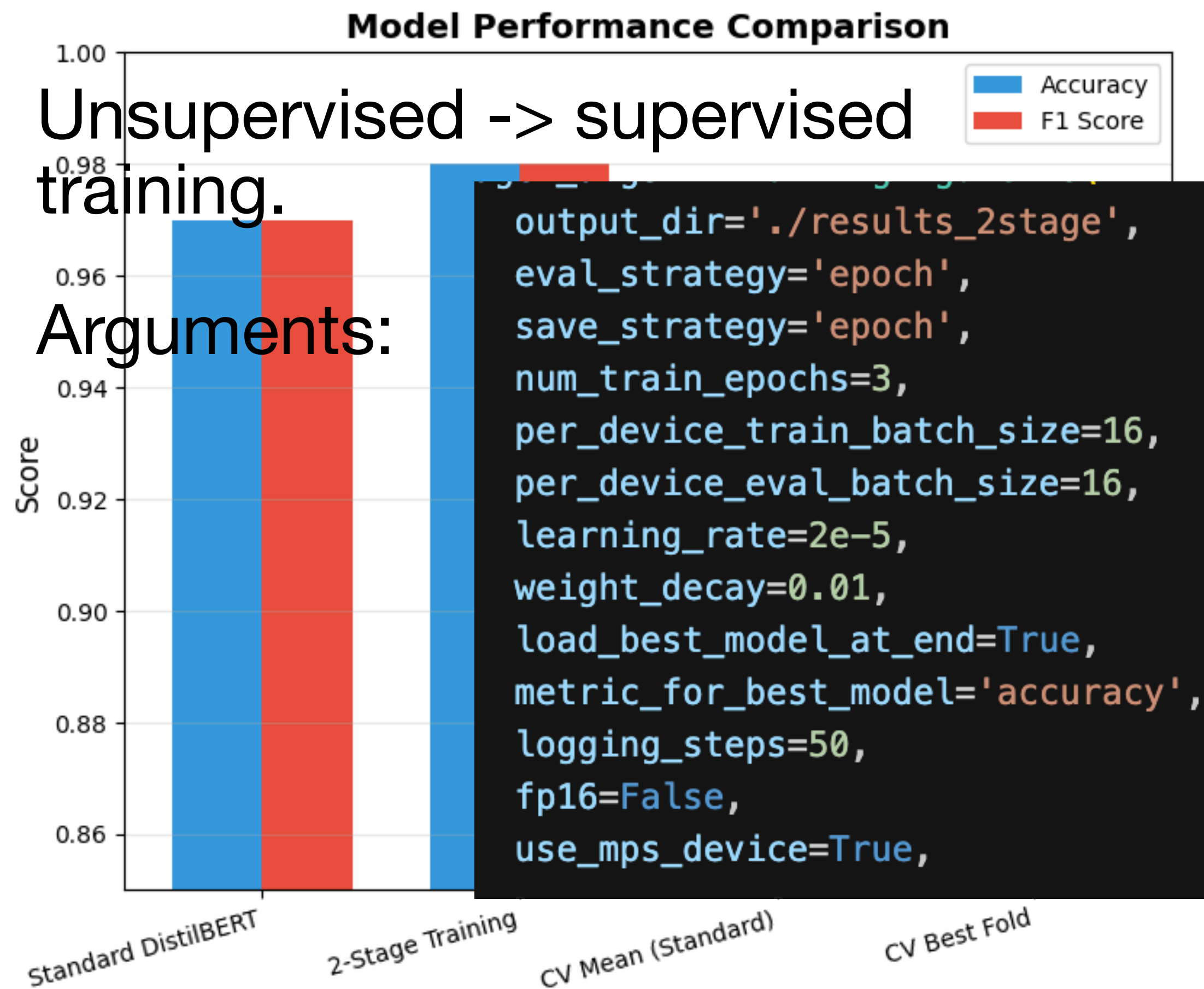
# Deep Learning Part

## DistilBERT: 2nd phase of fine-tuning

- Unsupervised -> supervised training.

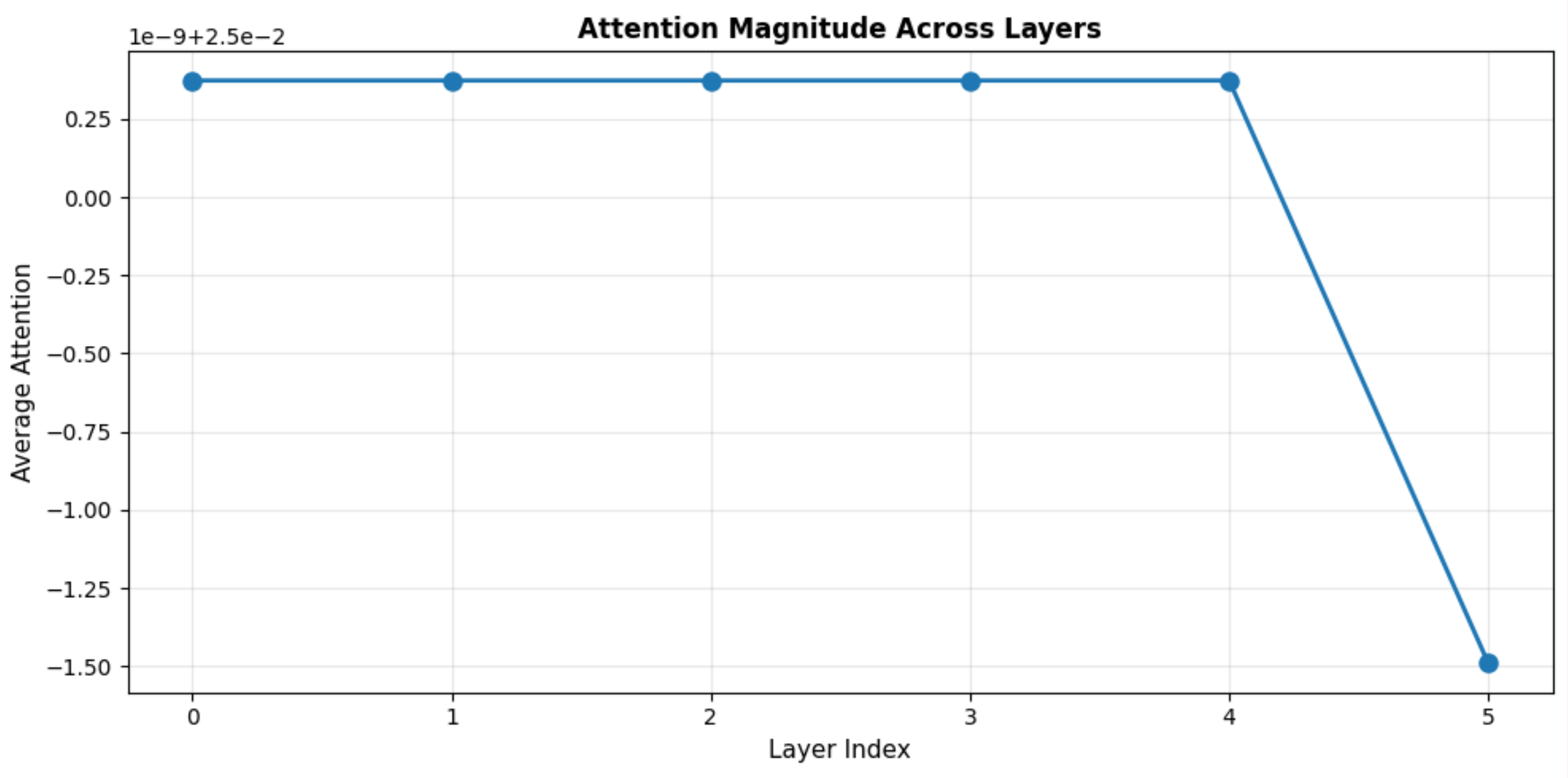
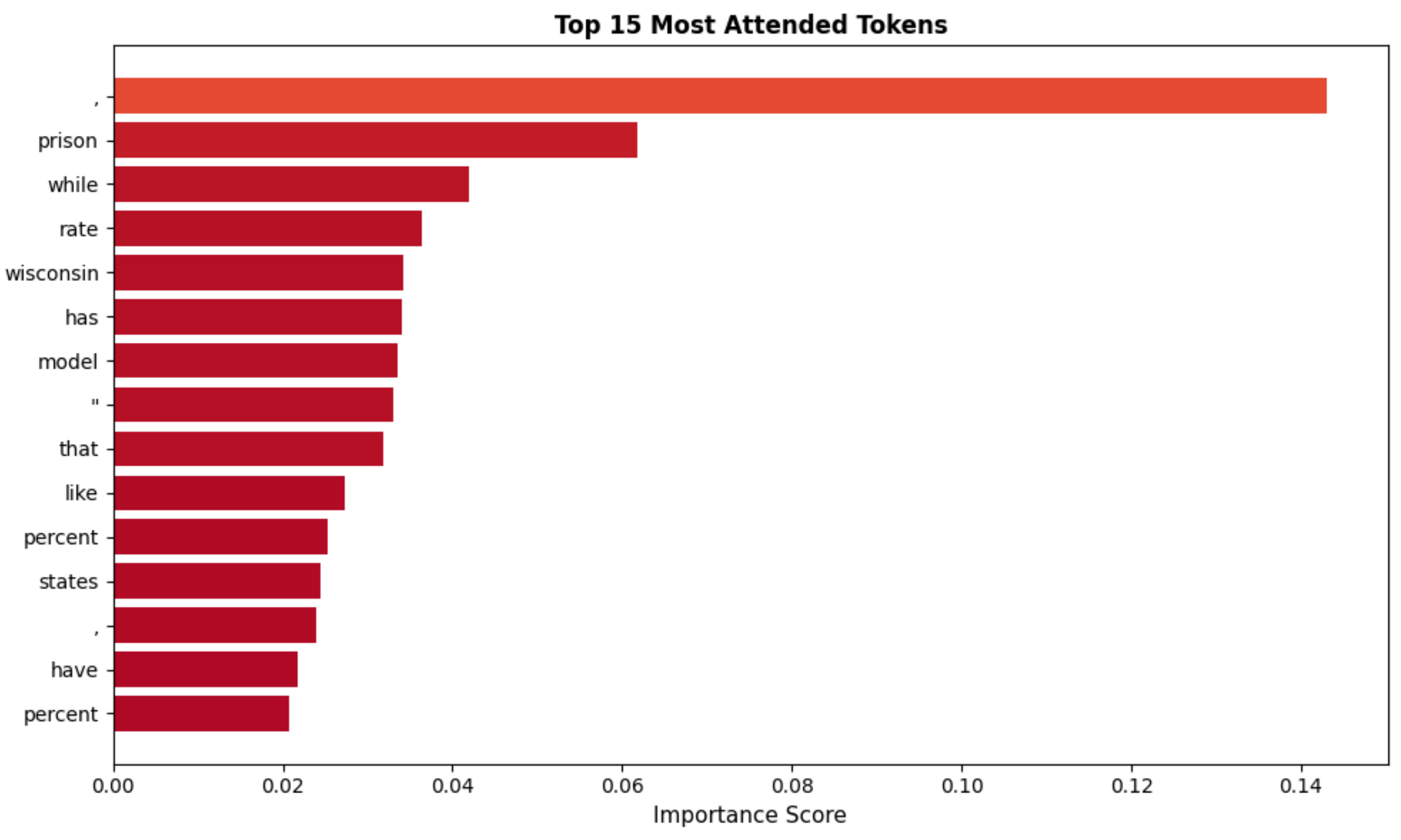
- Arguments:

```
output_dir='./results_2stage',  
eval_strategy='epoch',  
save_strategy='epoch',  
num_train_epochs=3,  
per_device_train_batch_size=16,  
per_device_eval_batch_size=16,  
learning_rate=2e-5,  
weight_decay=0.01,  
load_best_model_at_end=True,  
metric_for_best_model='accuracy',  
logging_steps=50,  
fp16=False,  
use_mps_device=True,
```





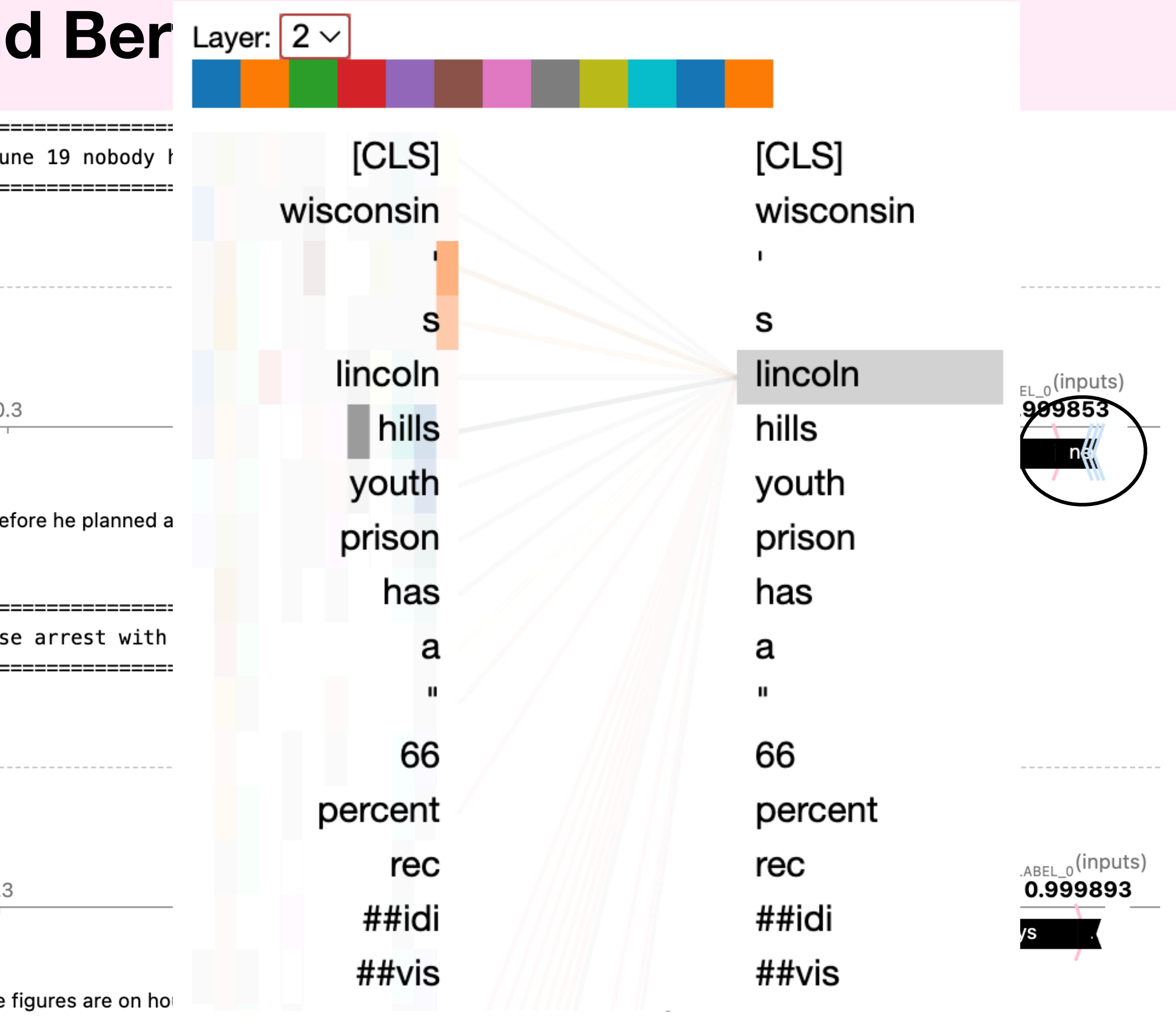
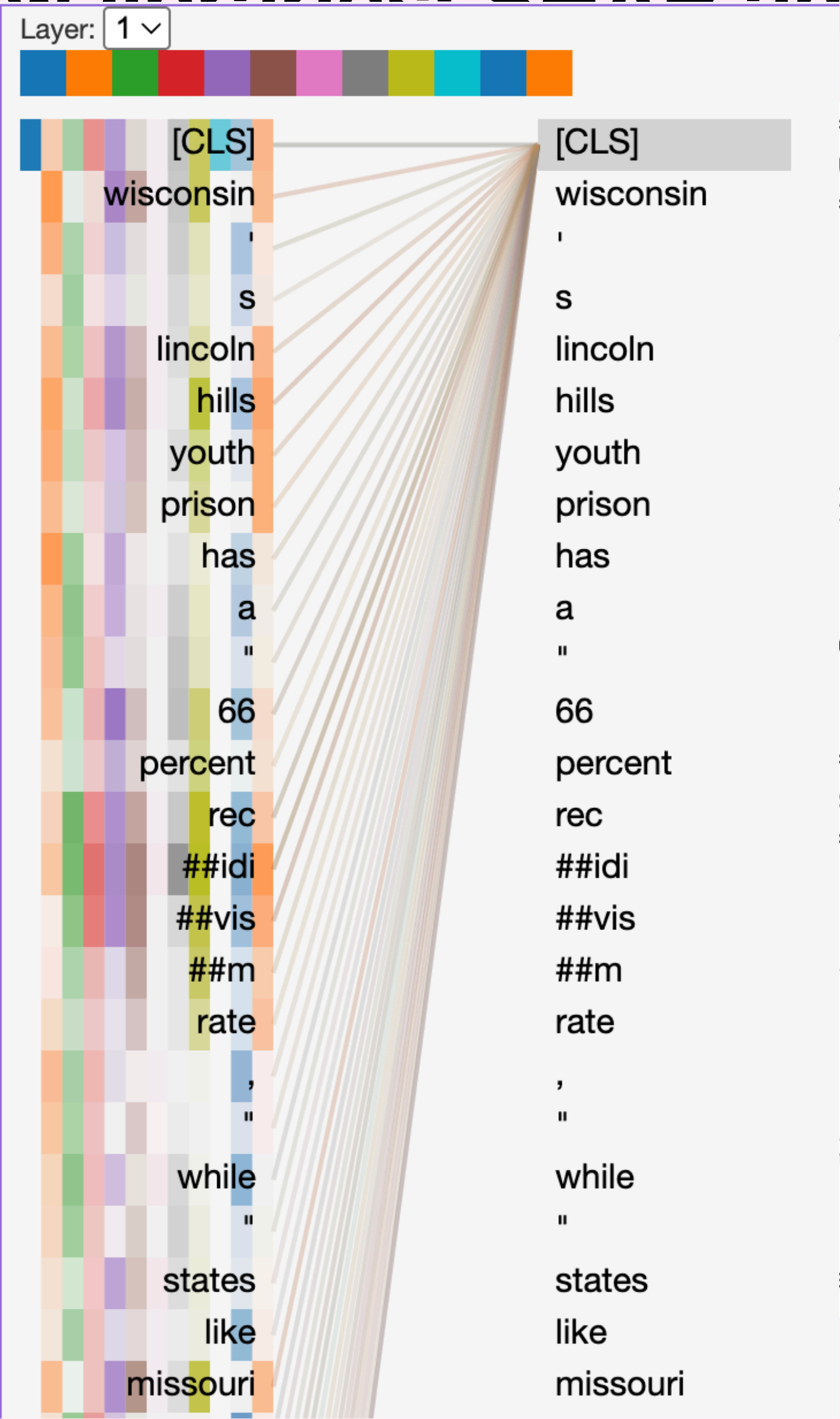
# DistilBERT: Dashboard





# DistilBERT: Dashboard

## New libraries: SHAD and Ber





# Deep Learning Part

## 6-Model Weighted Ensemble: Final Phase

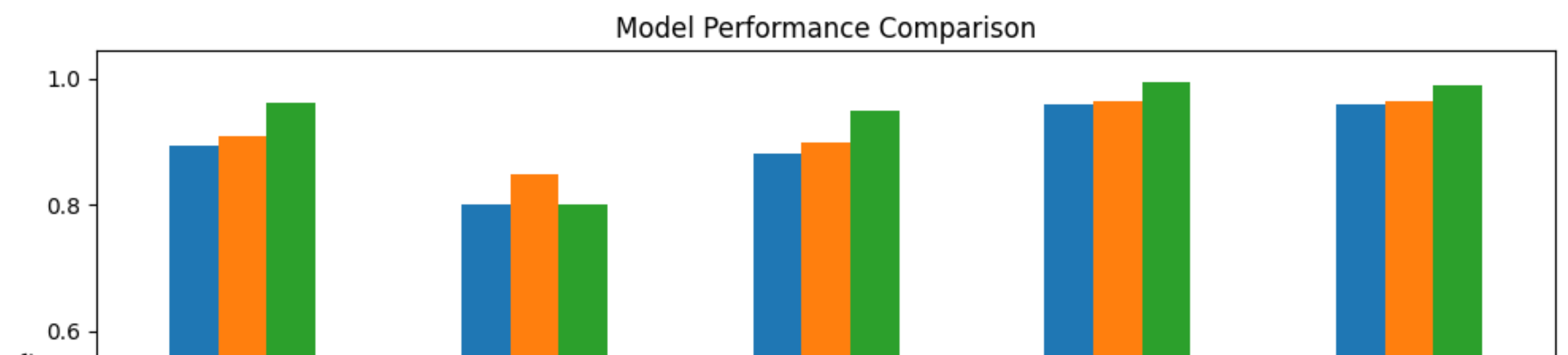
- Using the weights of each baseline model and transformer in order to create an ensemble model.

**DistilBERT (70%) + Gradient Boost(XGBoost (12%) + LightGBM (10%)) + Random Forest (5%) + Logistic Regression (3%).**

- Adding class weights to handle light imbalance:

(**FAKE** 43.9%): 1.1385 / (**TRUE** 56.1%): 0.8915





**1. A 2-stage approach:**

**Stage 1 (Masked Language Modeling):** DistilBERT on our specific news corpus (unlabeled) to teach it the "language of fake news" (vocabulary, style, sentence structure).

**Stage 2 (Classification):** then we fine-tune this "news-adapted" brain for the specific True/Fake task.

**2. Optimised Ensemble Weights:**

**Weights were not guessed but empirically optimised on validation performance:**

**- DistilBERT (70%):** The heavy lifter for semantic understanding. / **XGBoost (12%) & LightGBM (10%):** Catching non-linear patterns the transformer missed. / **Random Forest (5%) & LogReg (3%):** Providing stability and a linear baseline.

| Model               | Accuracy | F1-Score (Macro) | Precision (Avg) | Recall (Avg) | Training Time |
|---------------------|----------|------------------|-----------------|--------------|---------------|
| LightGBM            | 87.87%   | 0.8757           | 0.8811          | 0.8727       | ~10 min       |
| Logistic Regression | 88.00%   | 0.88             | 0.88            | 0.88         | ~2 min        |
| Random Forest       | 86.49%   | 0.86             | 0.86            | 0.86         | ~15 min       |
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# Part V Deployment with Streamlit



**Claim / Title article / News article**



**Score from ensemble model**  
**Comment from Google FactCheck Tools**  
**Justification by LLM**

**Translation / Multilingual UI based in IP /  
Text extraction**

**Chat & Debate**

**3 different LLMs**

**Compare 2 LLM Models**

**Deep Analysis**

**History Analysis /  
MLFlow Dashboard**



## Improvements:

1. It swings mostly from True to Fake, since it was trained on a two-class dataset. It needs to learn nuances.
2. Less confusing UI.
3. Tools like this need to be updated every day with new quality material, since information gets old very quickly these days. *Daily routine of web-scrapping and training?*
4. *Including media recognition (TikTok, YouTube, etc.).*
5. Integration directly into the browser (extension)/smartphone (app)





# Deployment with LLM APIs

## Short Guide

- Create an API key and save it as a *secret value* in your Docker or site.
- Choose the perfect deployment site (free [fly.io](https://fly.io) is not enough, GCR is too slow).
- Add one or more prompts for the LLM:
- Provide uni-test error for quota limit, API keys problems, prompts uncompleted...

[Hugging Face public repo](#)

[Hugging Face public model](#)

[Google Cloud Run repo](#)

```
system_prompt = "You are an expert, neutral fact-checking assistant. Be precise and cite sources."

user_prompt = (
    "You are a careful fact-checking teacher.\n"
    "Explain in 2-4 short paragraphs, accessible to non-experts, why the claim/article might be true or fake.\n"
    "Use external evidence if available. Keep balanced, avoid overclaiming, and cite sources as bullet links.\n\n"
    f"Claim/Article text (truncated):\n{_truncate(input_text)}\n\n"
    f"Model scores (probabilities): fake={model_scores.get('fake'):.3f}, true={model_scores.get('true'):.3f}.\n"
    f"Google Fact Check aggregate score (0-1): {google_score if google_score is not None else 'N/A'}.\n\n"
    f"External evidence:\n{evidence_str}"
)
```



# Advantages of using LLM

## **Local training/inference is free. API costs:**

- Google Fact Check (1000 free queries/day)
- OpenAI GPT (~\$0.01-0.03 per explanation)
- Perplexity (~\$0.01-0.05 per explanation)
- Gemini (~\$0.005-0.01 per explanation, FREE tier: 1500/day), BEST

## **All three provide LLM explanations:**

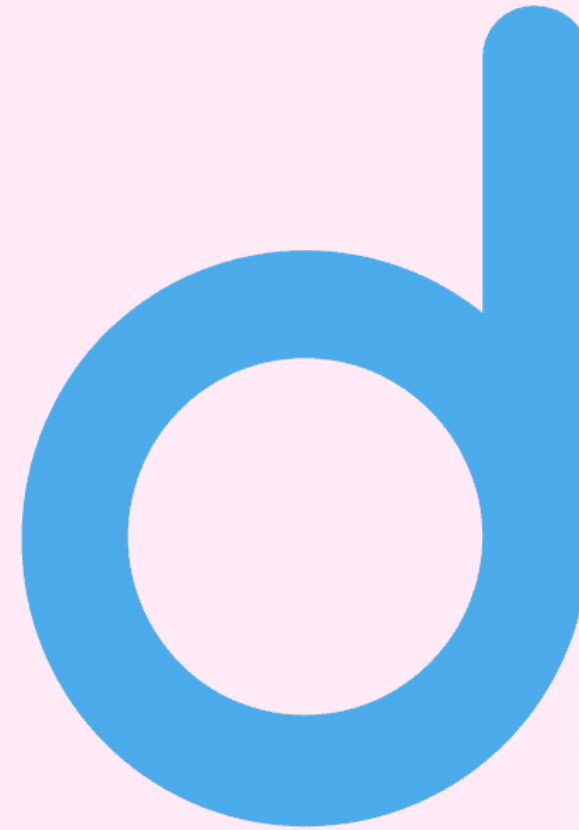
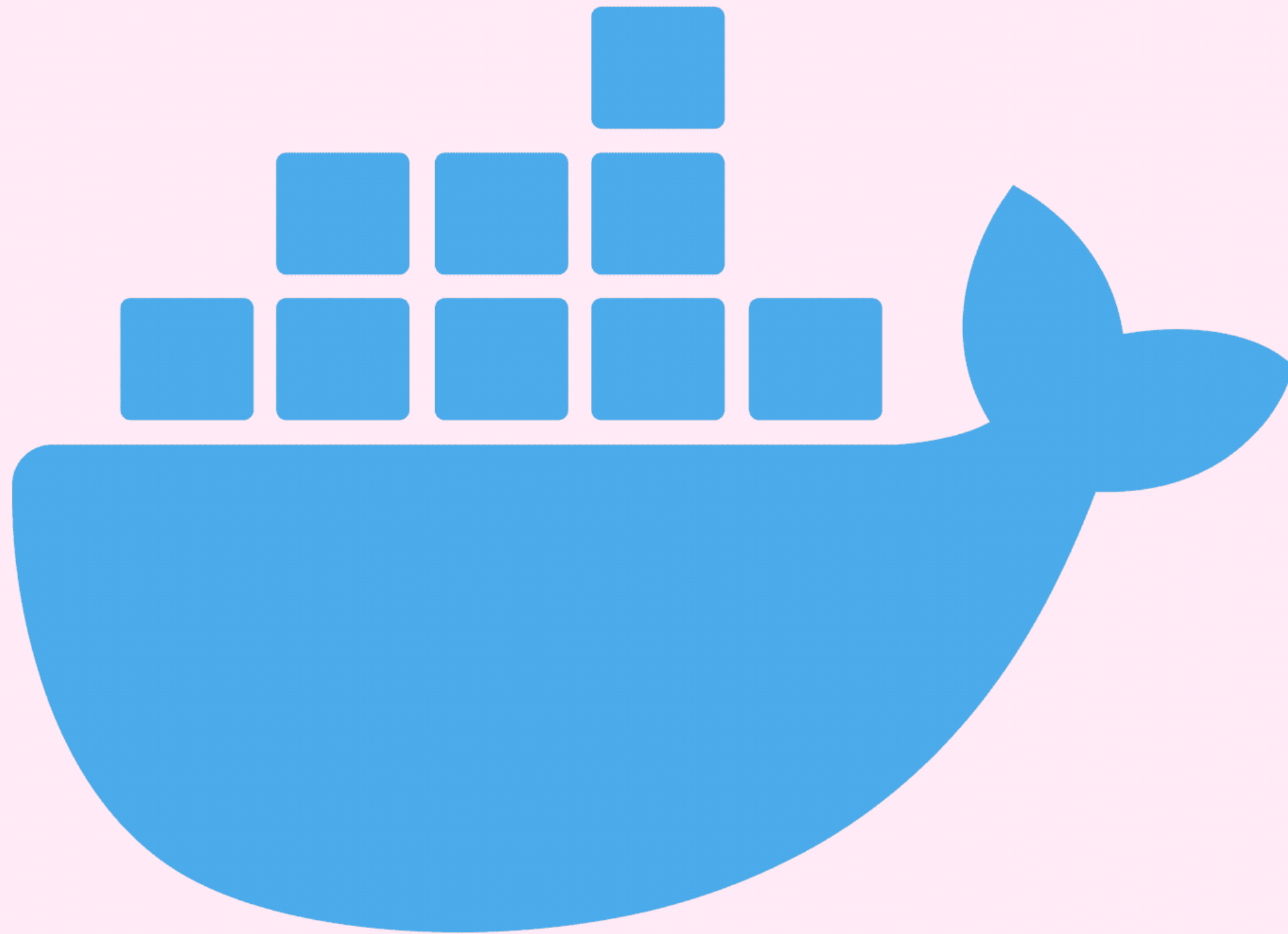
- OpenAI (GPT-4o-mini): Fast, reliable, well-documented. Best for production.
- Perplexity: Includes real-time web search. Best for current events.
- Gemini (2.5 Flash): Generous free tier (1500 req/day). Best for development and moderate usage.

## **Prompts ready to use for debate:**

- Why is true/fake?
- Show me evidence
- I disagree



# Docker and structure



```
FakeScope/  
├── README.md  
├── requirements.txt  
├── Makefile  
├── pytest.ini  
├── SUMMARY.md  
├── notebooks/  
│   ├── Project.ipynb  
│   ├── Development.ipynb  
│   ├── Other.ipynb  
│   └── Definition.ipynb  
├── datasets/  
│   ├── input/  
│   │   ├── alt/  
│   │   │   ├── News.csv  
│   │   │   └── fake_news_total.csv  
│   │   └── alt 2/  
│   │       └── New Task.csv  
├── models/  
│   ├── distilbert_news_adapted/  
│   │   ├── config.json  
│   │   ├── model.safetensors  
│   │   ├── tokenizer.json  
│   │   └── vocab.txt  
│   └── distilbert_fakenews_2stage/  
│       ├── config.json  
│       ├── model.safetensors  
│       ├── tokenizer.json  
│       └── vocab.txt  
├── mlm_results/  
│   ├── checkpoint-2868/  
│   ├── checkpoint-5736/  
│   └── checkpoint-8604/  
├── results_2stage/  
│   (gitignored)  
│   ├── checkpoint-2151/  
│   ├── checkpoint-4302/  
│   └── checkpoint-6453/  
├── results_fold{1-5}/  
│   (gitignored)  
│   ├── cv_metrics.json  
│   └── checkpoint-*/  
├── src/  
│   ├── __init__.py  
│   ├── config.py  
│   └── data_pipeline.py  
├── tests/  
│   ├── conftest.py  
│   ├── test_data_pipeline.py  
│   └── test_models.py  
├── scripts/  
│   └── __init__.py  
├── mlruns/  
├── docs/  
│   └── fakescope-complete.md  
├── tfidf_vectorizer.joblib  
├── best_baseline_model.joblib  
├── exists)  
├── cv_aggregate_results.json  
└── model_comparison_summary.json
```



# Part VI Testing and Implementation



# Unit-testing

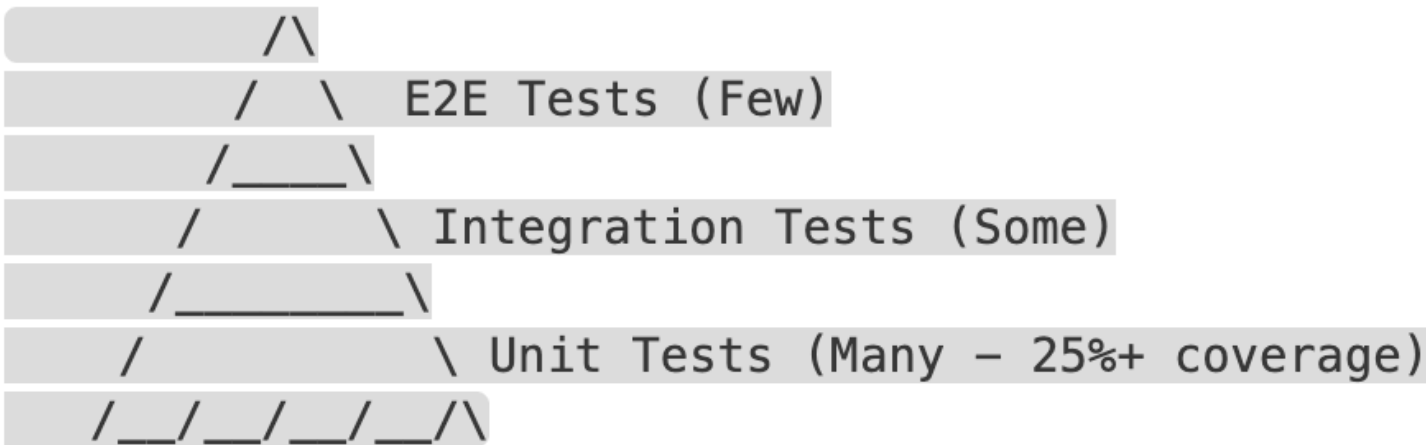
## Test Files

The project has comprehensive test coverage across 11 test files:

- `test_api.py` - API endpoint tests
- `test_data_pipeline.py` - Data loading & preprocessing
- `test_factcheck.py` - Google Fact Check API integration
- `test_inference.py` - Model inference logic
- `test_models.py` - Model configuration
- `test_openai_explain.py` - LLM explanation generation
- `test_shap_explain.py` - SHAP explainability
- `test_storage.py` - Database operations
- `test_translate.py` - Translation functionality
- `test_utils.py` - Utility functions
- `conftest.py` - Shared test fixtures

## Testing Strategy

### Testing Pyramid:



## Testing Framework

- **pytest**: Modern Python testing framework
- **pytest-cov**: Code coverage measurement
- **pytest-mock**: Mocking for isolated tests
- **hypothesis**: Property-based testing

## Coverage Goal: $\geq 25\%$

We'll focus on testing critical components:

1. Data preprocessing logic
2. Model training/prediction methods
3. Feature extraction
4. Input validation
5. Error handling

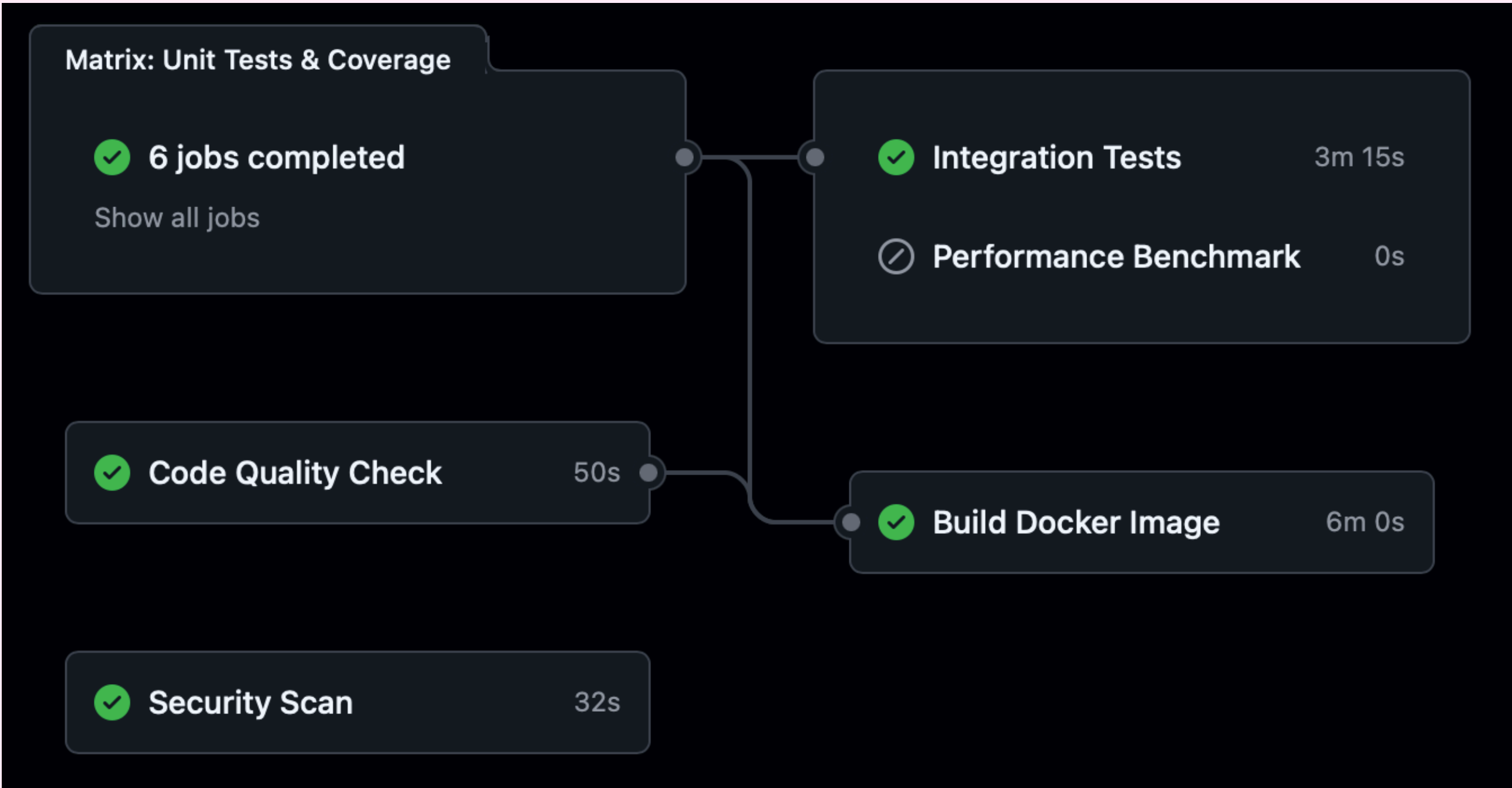
## Test Organization

```
tests/
├── __init__.py
├── conftest.py          # Shared fixtures
├── test_data_pipeline.py # Data loading & preprocessing tests
├── test_models.py       # Model training & prediction tests
├── test_features.py     # Feature extraction tests
└── test_integration.py  # End-to-end pipeline tests
```



# CD/CI Implementation

- ✓ Code Quality Check
- ✓ Unit Tests & Coverage (ubuntu-l...
- ✓ Unit Tests & Coverage (ubuntu-l...
- ✓ Unit Tests & Coverage (ubuntu-l...
- ✓ Unit Tests & Coverage (macos-l...
- ✓ Unit Tests & Coverage (macos-l...
- ✓ Unit Tests & Coverage (macos-l...
- ✓ Security Scan
- ✓ Integration Tests
- ✓ Build Docker Image





**THANK YOU,  
EVERYONE**

**I  DATA  
SCIENCE**

Enrique ESTEVEZ  
[https://github.com/enriest/  
FakeScope](https://github.com/enriest/FakeScope)